

SCQ (Single Correct Type) :

1. Ionic solid B^+A^- crystallizes in rock salt type of structure. 0.132 gm ionic solid salt B^+A^- is dissolved in water to make one litre solution. The pH of the solution is measured to be 6.0. If the length of face diagonal in the unit cell of B^+A^- be $600\sqrt{2}$ pm. Calculate the density of ionic solid. ($T = 298$ K), K_b for BOH is 10^{-5} . [Avogadro Number = 6.022×10^{23} , HA = strong acid]

(A) 5 g/cc (B) 4 g/cc (C) 4.5 g/cc (D) 5.5 g/cc

2. 10 ml of gaseous hydrocarbon is exploded with 100 ml oxygen. The residual mixture on cooling is found to measure 95 ml, of which, 20 ml is absorbed by KOH and remaining by alkaline pyrogallol. It is known that alkaline pyrogallol absorbs O_2 . Predict the formula of hydrocarbon :

(A) C_2H_6 (B) C_2H_4 (C) C_2H_2 (D) C_4H_8

3. At 70 K, the adsorption of N_2 gas at iron surface obeys Freundlich adsorption isotherm. The experimental data is :

P(bar)	4	25	64
$\frac{x}{m}$	0.2	0.5	0.8

Where x is the mass of N_2 in gram, adsorbed per gram of iron at P bar pressure. The mass of N_2 gas adsorbed by 5g of Fe at 36 bar and 70 K is :

(A) 0.6 g (B) 3g
(C) 0.8 g (D) Cannot be found from given data

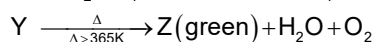
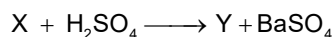
4. In which of the following cases, the work done in an ideal gas system with constant moles 'n' be represented by $\square nR\Delta T$?

(A) Isochoric reversible heating. (B) Isobaric reversible expansion
(C) Isothermal reversible expansion. (D) Adiabatic irreversible expansion.

5. A monoatomic ideal gas at 1200 K is expanded adiabatically from 1 L to 8L, such that its final temperature is 280 K. Which of the following statements is/are true for this process ?

(A) This is free adiabatic expansion. (B) This is reversible adiabatic expansion
(C) This is irreversible adiabatic expansion (D) This is an impossible process

6. $Cl_2(g) + Ba(OH)_2 \xrightarrow{\Delta} X(aq.) + BaCl_2 + H_2O$



Y and Z are respectively.

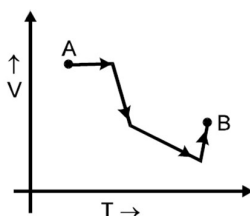
- (A) HClO_4 , ClO_2 (B) HClO_3 , ClO_2 (C) HClO_3 , Cl_2O_6 (D) HClO_4 , Cl_2O_7

7. Manganous salt in presence of catalyst zinc sulphate or zinc oxide is oxidised by potassium permanganate in neutral or faintly alkaline medium to :

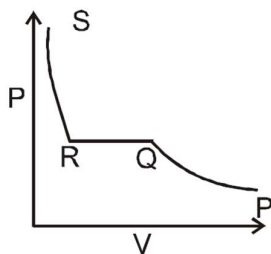
- (A) MnO_2 (B) Mn_2O_7 (C) Mn_2O_3 (D) Can not be oxidised

MCQ (Multiple Correct Type) :

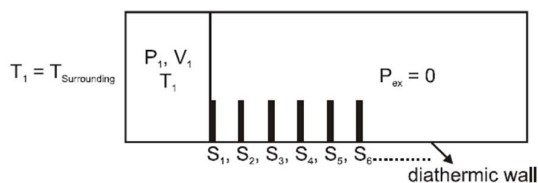
8. For the following V-T plot for a gas undergoing a process from state A to state B, select the correct alternative(s).



- (A) Pressure constantly increases (B) Pressure first increases, then decreases
(C) Final pressure is more than initial pressure (D) Pressure first decreases then increases.
9. The above isotherm was observed for a monoatomic gas at certain temperature. Which of the following is/are correct?



- (A) The gas is behaving ideally
(B) The gas is below its Boyle's temperature
(C) In the horizontal QR, the pressure is more than critical pressure
(D) The gas shows negative deviation from ideal gas in PQ and QR.
10. Which of the following statements is/are correct?
- (A) Higher is the gold number of a lyophilic sol, lesser is its coagulating power.
(B) An electrolyte can coagulate both positive sol and negative sol.
(C) Impurities present in a colloid makes it more stable.
(D) Dialysis is a process to remove impurities of ions and molecules from a colloid.
11. Consider the process carried out for an ideal gas.



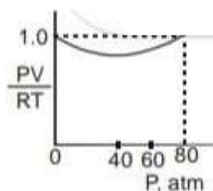
One by one, stoppers $S_1, S_2, S_3, \dots$ are removed slowly and the gas expands to V_2, L with final pressure P_2 and final temperature T_2 . Which of the following is/are correct for the above process.

- (A) $T_1 = T_2$ (B) $P_1 = P_2$ (C) $U_1 = U_2$ (D) $w = 0, q = 0$

12. Which of the following statement(s) is/are **false**?

- (A) $\Delta_r S$ for $\frac{1}{2}\text{Cl}_2(\text{g}) \rightarrow \text{Cl}(\text{g})$ is positive
 (B) $\Delta E < 0$ for combustion of $\text{CH}_4(\text{g})$ in a sealed container with rigid adiabatic system
 (C) ΔG is always zero for a reversible process in a closed system
 (D) ΔG° for an ideal gas reaction is a function of pressure

13. The Vander waal's equation of state for a non-ideal gas can be rearranged to give $\frac{PV}{RT} = \frac{V}{V-b} - \frac{a}{VRT}$ for 1 mole of gas. The constants a & b are positive numbers. When applied to H_2 at 80K, the equation gives the curve as shown in the figure. Which one of the following statements is(are) correct :



- (A) The two terms $\frac{V}{(V-b)}$ and $\frac{a}{VRT}$ are never equal
 (B) At 80 atm, the two terms $1 + \frac{V}{(V-b)}$ & a/VRT are equal.
 (C) At a pressure greater than 80 atm, the term $V/(V-b)$ is greater than $\frac{a}{VRT}$.
 (D) At 60 atm, the term $V/(V-b)$ is smaller than $1 + \frac{a}{VRT}$.

14. One gas bleaches the colour of flowers by reduction, while the other by oxidation, the two gases may be :

- (A) Cl_2 (B) H_2S (C) SO_3 (D) SO_2

15. Which of the following statements is/are correct about the reaction between the copper metal and concentrated HNO_3 ?

- (A) The principal reducing product is NO gas.

- (B) Cu metal is oxidised to Cu^{2+} (aq.) ion which is blue in colour.
 (C) All HNO_3 used act as oxidising agent.
 (D) The principal reducing product is NO_2 gas.
16. Aqueous solution of borax is treated with salicylic acid. Which of the following statements is/are **incorrect** for the product formed in the above reaction ?
 (A) no product will be formed because both are acids.
 (B) product is 4-coordinated complex and optically resolvable.
 (C) product is 4-coordinated complex and optically non-resolvable.
 (D) there are two newly formed five membered rings.
17. Select the correct statement(s) regarding reaction of SO_2 with PCl_5 .
 (A) It is a redox reaction.
 (B) One of the product is sulphuryl chloride.
 (C) Both the products on addition of water produce strongly acidic solutions.
 (D) Both the products have same hybridisation of central atom.
18. Which of the following statements is/are true?
 (A) In 3d series, there is a regular increase in the first ionisation enthalpy of transition elements from left to right.
 (B) In 3d series, the negative value of standard electrode potential (E°) for M^{2+}/M decreases in the order $\text{Ti} > \text{Mn} > \text{Cr} > \text{Fe}$.
 (C) The decrease in metallic radius coupled with increase in atomic mass results in a general increase in the density of transition elements from left to right
 (D) The higher oxidation state are favoured by the heavier elements (i.e. heavier members) in the groups of d-block.
19. Which of the following statements is/are true about Mohr's salt ?
 (A) It decolourises KMnO_4
 (B) It gives NH_3 on heating with KOH .
 (C) It is a double salt
 (D) Oxidation state of iron is +3 in this salt

Paragraph (One or more than one correct options) Que. 20 to 21

Zeta potential is a scientific term for electrokinetic potential in colloidal dispersions. It is usually denoted by using Greek letter zeta (ζ) It is the potential difference between dispersion medium and the stationary layer of fluid attached to the dispersed particle

20. What does a high value of zeta potential indicate :
 (A) High degree of repulsion between colloidal particles and medium
 (B) High degree of repulsion between colloidal particles
 (C) Easy coagulation
 (D) Electrically stable colloid.
21. Zeta potential approaches zero
 (A) in acidic medium (B) basic medium (C) at $\text{pH} = 7$ (D) at pH of isoelectric point

Paragraph (One or more than one correct options) Que. 22 to 23

Minerals having the formula AB_2O_4 and crystallizing in cubic system with O^{2-} ion forming fcc and A^{2+} & B^{3+} occupying octahedral & tetrahedral sites are generally termed as spinels.

Spinel is mainly two types : normal and inverse. In normal spinels, A^{2+} occupy one tetrahedral void and B^{3+} occupy two octahedral voids per fcc unit cell of O^{2-} . But in inverse spinel, A^{2+} occupy 1 octahedral void and B^{3+} occupy one tetrahedral and one octahedral void per fcc unit cell of O^{2-} ion.

22. $MgAl_2O_4$ is having normal spinel structure. Mg^{2+} and Al^{3+} are isoelectronic then which of the following statements is/are correct for $MgAl_2O_4$?

(A) Mg^{2+} being larger in size fits in octahedral void
(B) Al^{3+} being larger in size fits in octahedral void
(C) Al^{3+} occupy tetrahedral and octahedral void equally
(D) Mg^{2+} occupy tetrahedral void irrespective of the larger size than Al^{3+} .

23. Some spinels have distribution of A^{2+} and B^{3+} ions in tetrahedral and octahedral voids. If the formula is

$$(A_{1-x}B_x)\left(A_xB_y\right)_2O_4 \text{ then } y \text{ is:}$$

(A) 1 (B) X (C) $1 - \frac{x}{2}$ (D) $2 - x$

24. Match the following :

Column-I

(A) $Na_3[Co(ox)_3]$
(B) $[RhCl(CO)(PPh_3)(NH_3)]$
(C) $K[Cr(NH_3)_2(CN)_4]$
(D) $Ba[CoBr_2Cl_2]$

Column II

(p) Show geometric isomerism.
(q) Low spin complex.
(r) Paramagnetic with more than two unpaired electrons.
(s) Show colligative property corresponding of two particles per formula.
(t) Electrically conducting.

25. KCl has NaCl type face centred cubic crystal structure and CsF has CsCl type cubic crystal structure. Calculate the ratio of densities of CsF and KCl. It is given that the molar mass of CsF is twice that of KCl and edge length of KCl unit cell is 2 times that for CsF.

ANSWER KEY

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|-----|--|-----|-------|-----|-------|-----|---------|-----|---------|-----|---------|
| 1. | (B) | 2. | (C) | 3. | (B) | 4. | (B) | 5. | (D) | 6. | (B) |
| 7. | (A) | 8. | (B C) | 9. | (B D) | 10. | (B D) | 11. | (A C D) | 12. | (B C D) |
| 13. | (A C D) | 14. | (A D) | 15. | (B D) | 16. | (A C D) | 17. | (C D) | 18. | (B C D) |
| 19. | (A B C) | 20. | (B D) | 21. | (D) | 22. | (D) | 23. | (A) | | |
| 24. | (A $\rightarrow$ q , t) ; (B $\rightarrow$ p, q) ; (C $\rightarrow$ p, r, s, t) ; (D $\rightarrow$ r , s , t) | | | | | | | 25. | 4 | | |